

Application of machine learning to MRI, electroencephalography and brain computer interfaces

MRI Class 2

Diffusion MRI

1 Fundamentals of Diffusion MRI

Diffusion MRI (dMRI) is a powerful imaging technique used to measure the displacement of water molecules in biological tissues. This allows researchers and clinicians to infer microstructural properties, particularly in the brain. The sensitivity to movement is achieved using magnetic field gradients, which enable the detection of water molecule motion at a microscopic scale.

A key principle of dMRI is phase accumulation. Water molecules accumulate phase differences due to their movement between the RF pulse and the recording of the MRI signal. If no movement occurs, these phase differences cancel out, but when displacement is present, phase differences persist, leading to a measurable signal loss. This signal loss is proportional to the amount of molecular displacement, forming the foundation of diffusion MRI as a means of assessing tissue structure.

2 Diffusion Tensor Imaging (DTI) Model

A model that represents diffusion as a tensor, capturing directionality in multiple orientations.

A model of the local dMRI (in a voxel) is diffusion tensor imaging (DTI), which represents the directional properties of diffusion using a tensor. The diffusion tensor, a 3×3 matrix, captures both the magnitude and orientation of diffusion in different directions. When diffusion is isotropic, meaning movement is equal in all directions, the tensor takes on a spherical shape. In contrast, anisotropic diffusion, where movement is preferential along certain axes (as in white matter), results in an ellipsoidal tensor.

The diffusion tensor is characterized by its eigenvalues and eigenvectors. The eigenvalues ($\lambda_1, \lambda_2, \lambda_3$) quantify diffusion magnitude along principal directions, while the eigenvectors provide orientation information. In white matter, diffusion is highly directional due to the alignment of axonal fibers, leading to one dominant eigenvalue and a cigar-shaped tensor. In cerebrospinal fluid (CSF), where diffusion is unrestricted, all three eigenvalues are similar, resulting in a spherical tensor.

To measure the diffusion signal, MRI scanners acquire multiple images with different diffusion-weighted gradients. The signal is modelled as

$$S(g) = S_0 e^{-bg^T D g} \quad (1)$$

where S represents the measured signal, S_0 is the baseline signal without diffusion weighting, b is the B-value (which controls sensitivity to diffusion), g is the diffusion gradient direction, and D is the diffusion coefficient derived from the tensor model. At least seven different acquisitions are necessary to estimate the diffusion tensor accurately: six with diffusion weighting in different orientations and one baseline image (B_0).

3 Limitations of the Diffusion Tensor Model

Despite its usefulness, DTI has significant limitations. One major challenge is its inability to resolve complex fiber geometries within a single voxel. The model assumes a single dominant fiber orientation per voxel, which can lead to incorrect estimates in brain regions where multiple fiber bundles cross. For example, if two fiber bundles intersect perpendicularly, DTI may erroneously suggest isotropic diffusion, even though the underlying structure is more complex.

To overcome these limitations, advanced diffusion models have been developed. Techniques such as high-order spherical harmonics, diffusion spectrum imaging (DSI), Q-ball imaging (QBI), and constrained spherical deconvolution (CSD) enable more accurate reconstruction of fiber orientations in regions with crossing fibers.

4 Machine Learning for Tissue Classification Using dMRI

One practical application of dMRI is tissue classification, where different tissue types (white matter, gray matter, and CSF) are segmented based on their diffusion properties. Traditional approaches rely on thresholding diffusion metrics, but machine learning methods offer improved accuracy and robustness.

Deep learning, particularly convolutional neural networks (CNNs), can take diffusion-weighted images as input and classify voxels based on tissue type. Instead of using raw diffusion signals, machine learning models often rely on tensor-derived features such as mean diffusivity (MD), which represents the overall diffusion rate; fractional anisotropy (FA), which measures diffusion anisotropy; and the individual eigenvalues of the diffusion tensor. These features provide rich information about tissue microstructure, enabling the model to distinguish between white matter, which has high FA due to its structured fibers, and gray matter, where FA is lower due to more random diffusion.

5 Invariance in Machine Learning Models

When designing machine learning models for dMRI analysis, it is interesting to consider invariance properties. A well-constructed model should be invariant to transformations that do not affect the underlying biological structures. For example, in image classification, rotation invariance ensures that a neural network recognizes objects regardless of orientation. Similarly, in dMRI, models can focus on the eigenvalues of the diffusion tensor rather than the eigenvectors, which change with orientation.

Various strategies exist for enforcing invariance in machine learning models. One approach is architectural invariance, where neural network architectures are explicitly designed to disregard irrelevant transformations. CNNs, for instance, naturally enforce translation invariance through convolutional filters. Another method is data augmentation, where synthetic variations of training data are generated by applying transformations such as rotations or scaling, ensuring that the model learns to generalize across these variations. Additionally, feature transformation techniques can be employed, where input data is preprocessed into an invariant representation before being fed into the model. In the case of dMRI, this could involve computing rotation-invariant tensor features such as eigenvalues.

6 Practical Applications and Future Directions

Diffusion MRI has numerous applications in both research and clinical settings. One of its most significant uses is in brain connectivity mapping, where white matter fiber pathways are reconstructed through tractography. This technique is essential for understanding brain connectivity and diagnosing neurological disorders such as Alzheimer's disease, multiple sclerosis, and traumatic brain injury.